



CityShield Platform

Simulating and Defending Smart City Cyberattacks

University

University of Ha'il

Department

Information Security

Course

Graduation Project I [ISEC470]

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Key Objectives

- Modular smart city cyber range
- Realistic attack simulation
- Centralized logging & monitoring
- Automated response mechanisms

Introduction to CityShield

What happens when the digital systems that run our cities come under attack?



Smart Cities: Our Digital Reality

- ✓ **Digital integration** into daily infrastructure
- ✓ **Connected systems** powering modern lives
- ✓ **IoT devices** controlling critical services



Our Challenge

- ! **Limited training grounds** for defenders
- ! **Outdated detection methods** for modern threats
- ! **Real-world risks** of testing in production

Introduction to CityShield



The Hidden Vulnerability

- ⚠️ **Expanded attack surface** through IoT
- ⚠️ **Complex systems** with real-time constraints
- ⚠️ **Legacy systems** with modern vulnerabilities



Our Solution

- ★ **CityShield**: Safe environment for defense practice
- ★ **Real-world scenarios** without real-world risks
- ★ **Automated detection** and response capabilities

Smart City IoT Growth (2023-2025)

75% increase

Source: Gartner IoT Market Forecast, Q3 2025

Feasibility Analysis

"When we first gathered to discuss this project, we asked ourselves: Can a team of students actually build something this ambitious? The answer surprised even us."



Technical Feasibility

- ✓ **Containerization** (Docker) for isolation
- ✓ **Orchestration** (Kubernetes/Docker Compose)
- ✓ **Logging Stack** (OpenSearch, dashboards)
- ✓ **Attack Simulation** with community frameworks

Implementation Complexity: **Medium**



Cost-Benefit Feasibility

- ↓ **Low Infrastructure Costs** (modest hardware)
- ↓ **Open-Source Tools** (no licensing fees)
- ↑ **Educational Value** (hands-on training)
- ↑ **Reusable Platform** (future courses/research)

ROI: **High**

Feasibility Analysis



Organizational Feasibility

-  **Faculty Mentorship** available
-  **Student Skills** align with project needs
-  **No Proprietary Licenses** required

Support Level: **Strong**

Our supervisor's enthusiasm for this project has been a huge motivator for the team.



Schedule Feasibility

-  **Planning & Research:** 1-2 weeks
-  **Environment Setup:** 1-2 weeks
-  **Component Development:** 3-4 weeks
-  **Integration & Testing:** 2-3 weeks

Total Timeline: **10-16 weeks**

We've built in buffer time for unexpected challenges - a lesson learned from previous projects!

Requirements Analysis

"After countless discussions and debates, we realized that designing for three distinct user types would be the key to making CityShield truly effective."



Administrator

System setup, configurations, access control

"The unsung hero who keeps everything running smoothly"



Security Analyst

Monitors alerts, investigates suspicious activity

"Our frontline defender against cyber threats"



Researcher

Designs scenarios, evaluates detection logic

"The innovator pushing boundaries of what's possible"

Requirements Analysis



Functional Requirements



F1 Environment Provisioning: Deploy simulation components



F2 Attack Simulation: Configurable attack scenarios



F3 Logging & Monitoring: Centralize log data



F4 Threat Detection: Identify threat activity



F5 Automated Response: Predefined response actions



F6 Dashboards & Visualization: System status views



Non-Functional Requirements



NFR1 Scalability: Support growing simulation components



NFR2 Reliability: Continuous operation during simulations



NFR3 Usability: Intuitive interfaces for users



NFR4 Performance: Timely processing of logs and alerts



NFR5 Extensibility: Future addition of scenarios and modules

System Modeling



Use Cases



Configure Platform

Administrator sets up environment and manages configurations



Run Simulation

Researcher initiates and controls simulation scenarios



Monitor Events

Security Analyst observes activities through dashboard



Investigate Alerts

Security Analyst examines correlated events and details



Generate Reports

Researcher prepares evaluation summaries of outcomes



Sequence Diagrams



Simulation Start and Logging Flow

Researcher

Orchestrator

SimComp

LogCollector

Start Scenario

Deploy
Components

Generate Logs

Ingest Logs



Detection and Alert Triggering

LogCollector

DetectionEngine

AlertStore

Dashboard

Forward Logs

Evaluate Rules

Generate Alert

Notify Analyst

Database Design



Data Requirements



Log Events

Raw event data from simulation components



Alerts

Detection engine notifications for suspicious behavior



Detection Rules

Definitions used to identify suspicious patterns



Reports

Summaries of exercise outcomes and performance metrics



Scenario Metadata

Configurations and parameters for simulation scenarios



Design Principles



Searchability

Efficient log queries and filtering



Scalability

Ability to handle large volumes of events



Extensibility

Schema evolution as new event types are added



Analytics Support

Aggregations and dashboards without heavy ETL



Indexing Strategy

Index	Fields
logs_*	timestamp, source_host, event_type, message, metadata
alerts_*	alert_id, rule_id, severity, triggered_at, related log IDs
reports_*	report_id, scenario_id, summary, metrics, generated_on
scenarios_*	scenario_id, description, components, duration, attack_pattern
rules_*	rule_id, description, conditions, severity, actions

Architectural Design



System Architecture



UI Layer



Dashboards



Scenario Builder



Alert Panel



Detection Layer



Detection Engine



Rule Manager



Response Manager



Data Layer



Log Store



Alert Store



Analytics Engine



Simulation Layer



Traffic Sim



IoT Sensor Sim



Network Emulator



Component Descriptions



Simulation Services

Containerized components representing smart city subsystems



Logging Pipeline

Centralized log aggregation, indexing, and visualization



Detection Engine

Rule-based analysis to identify suspicious activity



Response Manager

Coordinates automated response actions based on alerts

Architectural Design



Security Considerations



Logical Isolation

Complete separation from production environments



Network Segmentation

Restricted communication paths between components



Controlled Access

Limited access to dashboards and analytical outputs



Secure Configuration

Container hardening and vulnerability management

User Interface Design



Main UI Modules



Overview Dashboard

Platform health, ingestion status, security summary



Alert & Investigation

Filter alerts, view details, pivot to logs



Scenario Builder

Define scenarios, select components, set attacks



Overview Dashboard



System Health

All systems operational
100%



Event Rate

Events per second
320



Critical Alerts

Active critical alerts
1



High Alerts

Active high alerts
3



Alert & Investigation Panel

Time	Severity	Rule ID	Component	Status
12:31	High	EXAMPLE_NET_SCAN	traffic_sim	Open
12:29	Medium	EXAMPLE_IOT_ANOM	iot_sensor_sim	Triaged
12:27	Low	EXAMPLE_AUTH_FAIL	auth_service	Resolved

User Interface Design



Scenario Builder Interface

Scenario ID

Description

Components

traffic_simulator (event_rate: 150)

iot_sensor_sim (anomaly_density: low)

log_collector (pipeline: default)

Attack Pattern

Similar Projects in the Market

Cyber Range Platforms Comparison

CityShield compared against commercial and academic cyber range solutions

Feature	Commercial CR	Academic Testbeds	CityShield
Smart City Focus	No	No	Yes
Scenario Customization	Medium	Variable	High
Enterprise SIEM Integration	Partial	Low	Strong
Automated Response	Partial	No	Yes
Research Extensibility	Medium	High	High



Seamless Integration

Native integration with **OpenSearch/Kibana** for logging and visualization, unlike many platforms that require custom connectors



Smart City Specific

Purpose-built for **smart city environments** with IoT, traffic systems, and urban infrastructure simulations



Educational Focus

Designed for **academic environments** with lower cost, open-source approach, and educational workflows

Log Aggregation Methods Comparison

→ Open Source vs Commercial Solutions

Analysis of leading log aggregation platforms for CityShield implementation

<> Open Source Solutions

ELK Stack

- ✓ Powerful search and visualization
- ✓ Active community support
- ✓ Flexible data ingestion
- ✗ Resource-intensive for large deployments

OpenSearch

- ✓ Fork of Elasticsearch with AWS backing
- ✓ Security features included by default
- ✓ No licensing restrictions
- ✗ Smaller community than ELK

Commercial Solutions

Splunk

- ✓ Advanced analytics and machine learning
- ✓ Enterprise-grade security features
- ✗ High licensing costs
- ✗ Complex deployment and maintenance

Datadog

- ✓ Excellent visualization capabilities
- ✓ Strong APM integration
- ✗ Subscription-based pricing model
- ✗ Less flexible for custom use cases

Log Aggregation Methods Comparison



Why Kibana for CityShield

-  **Cost-Effective:** No licensing fees, ideal for academic environments
-  **Extensible:** Rich plugin ecosystem for custom visualizations
-  **Easy Integration:** Native support for containerized environments
-  **Educational Value:** Widely used in industry, enhancing student skills



Feature Comparison

Feature	Kibana	Splunk	Datadog
Search Performance	Very Good	Excellent	Very Good
Visualization	Very Good	Very Good	Excellent
Cost Efficiency	Excellent	Poor	Average

Source: Gartner Magic Quadrant for SIEM Platforms, Q3 2025

MTTD and MTTR Comparison

Time Metrics

Comparing CityShield targets with industry benchmarks and best-in-class solutions



MTTD

Mean Time To Detect: Time from attack initiation to detection

Industry Average



Less than 5 hours

SANS 2023 SOC Survey: 50% of organizations had MTTD < 5 hours



Best-in-Class

60 minutes or less

SANS 2023 SOC Survey: 25% had MTTD < 60 minutes



CityShield Target

20 Minutes



MTTR

Mean Time To Respond: Time from detection to response execution

Industry Average



2-4 hours

LinkedIn: For MTTR, a two-to-four-hour range is generally acceptable across all alert severities



Best-in-Class

20 minutes

Organizations with automated response systems



CityShield Target

10 Minutes

Differentiators of CityShield Platform

★ What Makes CityShield Unique

Key advantages that set CityShield apart from existing cyber range solutions



Smart City Focus

Superior

- ✓ **Purpose-built** for urban infrastructure
- ✓ **IoT integration** with sensor simulations
- ✓ **Traffic system** modeling capabilities



Native Kibana Integration

Innovative

- ✓ **Seamless** log aggregation and visualization
- ✓ **Custom dashboards** for smart city metrics
- ✓ **Open-source** with no licensing constraints



Response Performance

Advanced

- ✓ **MTTD**: 20 Minutes (vs. industry 60 minutes)
- ✓ **MTTR**: 10 minutes (vs. industry around 20 minutes)
- ✓ **Automated** response orchestration



Educational Focus

Academic

- ✓ **Cost-effective** for academic institutions
- ✓ **Research-friendly** with extensible architecture
- ✓ **Industry-relevant** skills development

Conclusion



Key Achievements



Smart city focus with IoT and traffic simulations



Kibana integration for advanced visualization



Exceptional metrics: MTDD 20m, MTTR 10m



Educational value for cybersecurity training



Technical Foundation



Containerized microservices architecture



Centralized logging with efficient indexing



Rule-based detection with MITRE ATT&CK alignment



Intuitive UI for analysts and researchers



What's Next for CityShield



**Cloud-native
Deployment**



**Multi-user Training
Modules**



**Threat Intelligence
Integration**

Conclusion

Thank You

We're excited to share our work and answer your questions

— The CityShield Team